

TANVI GANESH JOSHI

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EDUCATION

University of California, San Diego

MS in Computer Science and Engineering – GPA: 3.97/4.0

San Diego, CA

Sept 2024 – Jun 2026

Indian Institute of Technology, Bombay

B.Tech. in Mechanical Engineering, Minor in CSE – CPI: 8.85/10.0

Mumbai, India

Nov 2020 – May 2024

EXPERIENCE

BeOne Medicines

June 2025 – Mar 2026

Advanced Analytics & ML Intern, Marketing Analytics

San Carlos, CA

- Built a **KOL identification engine** using graph network analysis over **1M+ HCP interactions** from 20+ sources, surfacing high-influence prospects to concentrate marketing spend across multiple oncology products.
- Drove media mix modeling and multi-channel marketing analytics across multiple signals; automated Dataiku workflows and coverage audits to optimize spend allocation and cut redundant third-party data spend by **~70%**.
- Built synthetic HCP persona agents simulating responses to marketing tactics, enabling A/B testing of messaging pre-deployment, and benchmarked vendor coverage to standardize on the most comprehensive data source.

Halıcıoğlu Data Science Institute, UCSD

Apr 2026 – Jun 2026

Graduate Teaching Assistant – Systems for Scalable Analytics (Prof. Umesh Bellur and Prof. Alexander Kolovos)

San Diego, CA

- Graduate TA for **240+ students**, covering distributed computing, MapReduce, Spark, and end-to-end ML at scale.
- Led discussions and exam prep and supported PAs, evaluating students on dataflow systems and scalable AI workloads.

American Express

May 2023 – July 2023

Data Scientist, Decision Science & Strategy Intern

Mumbai, India

- Built and productionized an end-to-end **propensity model** over **12M+** cardmember records to predict B2B-merchant transaction likelihood and select high-value, ad-targetable cohorts for offer targeting and prospecting.
- Engineered the B2B merchant universe: trained a **multi-class merchant classifier** (B2B/B2C/C2C) on transaction logs and derived spend- and card-type features for ground-truth labels; handled class imbalance via balanced sampling.
- Benchmarked XGBoost vs. decision tree on accuracy, F1, and operational complexity to quantify the precision–maintenance cost tradeoff; delivered **~0.70 F1** and **~2.5× lift**, with feature importance validating top revenue signals.

PROJECTS

Hierarchical Demand Forecasting, Price Elasticity & Causal Inference | Darts, LightGBM, XGBoost, PyMC, FAISS

2026

- Built a pipeline for supply-chain planning (4-level hierarchy, 39 series) across ARIMA, Prophet, LSTM, and XGBoost (Darts); MinTrace-WLS reconciliation cut product-level RMSE to **10.56**, beating top-down by **28–54%**.
- Estimated price elasticity via a Bayesian hierarchical model (PyMC), recovering true category elasticities within 90% intervals; simulated $\pm 10\%$ pricing scenarios for revenue-optimal moves (**+4.8% revenue**).
- Quantified promotion lift with difference-in-differences and A/B testing, recovering a **+11.5% causal lift** on 1M+ real transactions despite selection bias; seeded cold-start SKUs via FAISS similarity (**–11.3% RMSE**).

Customer Churn Prediction with Causal Attribution | XGBoost, SHAP, scikit-learn

2025

- Built an XGBoost churn classifier on the IBM Telco dataset (7K customers) with engineered domain features; achieved AUC-ROC 0.849, **68% precision**, and 2.55× lift at the top-20% risk tier.
- Segmented churners into 4 cohorts via K-Means in SHAP space; top 2 cohorts captured 75% of churn and \$257K of \$327K revenue at risk, enabling targeted retention projected to cut churn **18–22%**.

Two-Tower Neural Recommendation System | PyTorch, FAISS, Python

2026

- Built a two-stage neural retrieval & ranking pipeline on MovieLens 1M; dual-encoder with InfoNCE loss and FAISS IVFFlat indexing achieving 5× speedup and 0.006ms/query latency.
- Ranking MLP trained on hard negatives improved NDCG@10 by **110%** and MRR by **85%** over a retrieval-only baseline; evaluated with Recall@K, NDCG@K, and MRR.

Multi-Agent LLM System with RL-Based Belief Inference (Avalon) | Python, PyTorch

2026

- Built a multi-agent LLM simulation with role-conditioned prompting, structured memory, and RL-based belief inference (suspicion + role prediction), backed by a game engine enforcing partial-information state.
- Improved accusation accuracy to **0.73** and reached **0.83** accuracy for low-information agents via RL policy learning.

Biomedical RAG Pipeline with Evaluation | sentence-transformers, FAISS, BM25

2026

- Built an end-to-end biomedical RAG system over 492 PubMed NSCLC abstracts (dense FAISS retrieval, BM25, and Reciprocal Rank Fusion hybrid), benchmarking chunking × retrieval strategies.
- Hybrid RRF retrieval lifted **MRR from 0.51 to 0.83** and P@5 to 0.56 over a dense-only baseline; built a 20-query labeled benchmark scoring Precision@5, MRR, and faithfulness to quantify hallucination risk.

TECHNICAL SKILLS

Languages & Data & Systems: Python, SQL, Hive, Hadoop, Spark, Airflow, Dataiku, Ray, AWS (S3, EC2, Redshift), FAISS, REST APIs

Libraries: PyTorch, TensorFlow, Keras, Scikit-Learn, PyMC, Darts, Prophet, statsmodels, Pandas, NumPy, Transformers, LangChain

Focus Areas: NLP & LLMs, Recommender Systems, Forecasting & Time-Series, Predictive Modeling, Ranking & Targeting, Agentic AI

Miscellaneous: KVPY Fellow, ME119 TA (IITB), Dept Academic Mentor (IITB), Grad Student Mentor (UCSD, IITB)